



Advanced Terraform on Azure

Lab 3 – Challenge: Refactoring for Scale and Controlled Validation

Overview

You have been provided with a working Terraform configuration at `c:\qatfadvez\lab3\guided`.

The infrastructure deploys successfully, but parts of the implementation are:

- repetitive
- rigid
- not scalable

Your task is to refactor the configuration so that it aligns with the dynamic, data-driven patterns introduced previously.

This challenge is completed in two phases.

Phase 1 – Refactor for Scale

Objective

- data-driven
- scalable
- deterministic
- defensively validated

Current State

- defines subnets using repetitive resource blocks
- requires Terraform changes when new subnets are added
- has no subnet-to-NSG associations configured
- does not sufficiently protect against bad subnet input

Your Task

- subnets are to be generated dynamically from input data
- no individual subnet resources should remain
- subnet-to-NSG associations are to be introduced
- associations are to be created automatically for each subnet
- the design should be stable and predictable
- invalid subnet CIDR input should be rejected early

Constraints

- do NOT implement Azure-specific subnet restrictions
- do NOT use the `approved_vnet_cidrs` variable
- do NOT use the supplied validation tfvars files
- focus on Terraform design and generic validation only

Success Criteria

- Terraform applies successfully
- subnets are fully data-driven
- no hardcoded subnet resources remain
- NSG associations are dynamic
- reordering input results in no changes
- adding a subnet results in only new required resources
- malformed subnet CIDR input fails during terraform plan

Transition – Enabling Governance Constraints

Before beginning Phase 2:

Uncomment the `approved_vnet_cidrs` variable in `variables.tf`.

This enables additional validation logic required for platform-specific constraints.

Phase 2 – Enforce Platform Constraints (Stretch)

Objective

Extend your Phase 1 solution to enforce Azure-specific network constraints.

Your Task

- `vnet_address_space` must be a valid CIDR and match one of the approved values defined in `approved_vnet_cidrs`
- all subnets must be at least /29 in size (Azure minimum)

- all subnets must be smaller than the VNet (subnet prefix length must be greater than the VNet prefix length)
- NSG rule priorities must be unique per direction (inbound and outbound rules must each have unique priorities)

Validation Requirements

- good.tfvars must succeed
- invalid tfvars must fail during terraform plan

Solution

Proposed solutions can be found at

- c:\qatfadvaz\lab3\solution\challenge_phase1_solution
- c:\qatfadvaz\lab3\solution\challenge_phase2_solution

For post-course use, a 'step-by-step' guided version of this lab can be found at:

- c:\qatfadvaz\lab3\solution\Lab3-Guided-v1.0.pdf

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